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CARRERA DE INGENIERÍA INFORMÁTICA

Curso: SISTEMAS OPERATIVOS

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ACTIVIDAD 1: Observando procesos en Linux

Objetivo: Identificar procesos reales y su comportamiento.

Comandos:

- top

```
i:Def - 18:32:08 up 13 min, 1 user, load average: 0.05, 0.10, 0.24
Tasks: 273 total, 1 running, 272 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.1 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 30131.4 total, 19707.3 free, 1866.3 used, 9265.6 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 28265.0 avail Mem
```

1	PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
	3802	ubuntu	20	0	6132552	400688	173992	S	1.3	1.3	0:37.88	gnome-shell
	195	root	20	0	0	0	0	I	0.3	0.0	0:01.13	kworker/u28:1-events_power_efficient
	7238	ubuntu	20	0	3706720	261148	147176	S	0.3	0.8	0:02.67	ptxixis
	1	root	20	0	26364	17588	11896	S	0.0	0.1	0:02.64	systemd
	2	root	20	0	0	0	0	S	0.0	0.0	0:00.04	kthreadd
	3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
	4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp

2	PID	PPID	TIME+	%CPU	%MEM	PR	NI	S	VIRT	RES	UID	COMMAND
	7313	2	0:00.00	0.0	0.0	20	0	I	0	0	0	kworker/4:0-events
	7298	7287	0:00.23	0.0	0.0	20	0	R	22204	6164	1000	top
	7287	7238	0:00.02	0.0	0.0	20	0	S	19908	5476	1000	bash
	7238	7230	0:00.04	0.0	0.0	20	0	S	240804	7840	1000	ptxixis-agent
	7230	3209	0:02.67	0.3	0.8	20	0	S	3706720	261148	1000	ptxixis
	7199	2	0:00.00	0.0	0.0	20	0	I	0	0	0	kworker/u25:0-events_power_efficient
	6865	3209	0:00.38	0.0	0.2	20	0	S	1164832	78092	1000	gnome-calendar

3	PID	%MEM	VIRT	RES	CODE	DATA	SHR	nMaj	nDRT	%CPU	COMMAND
	3802	1.3	6132552	400688	8	479984	173992	20k	0	1.3	gnome-shell
	7238	0.8	3706720	261148	216	461816	147176	1	0	0.3	ptxixis
	4958	0.4	1720480	118968	12	201128	94816	0	0	0.0	mutter-x11-frame
	3165	0.3	739888	101804	2956	143436	26616	83	0	0.0	python3.12
	5268	0.2	374008	72172	68	29992	65444	0	0	0.0	ibus-x11
	6572	0.2	3275788	71640	8	94280	55224	2	0	0.0	gjs
	4574	0.2	212344	71076	1756	8464	62808	2	0	0.0	Xwayland

4	PID	PPID	UID	USER	RUSER	TTY	TIME+	%CPU	%MEM	S	COMMAND
	3209	1	1000	ubuntu	ubuntu	?	0:00.65	0.0	0.0	S	systemd
	3218	3209	1000	ubuntu	ubuntu	?	0:00.00	0.0	0.0	S	(sd-pam)
	3247	3209	1000	ubuntu	ubuntu	?	0:00.03	0.0	0.0	S	dbus-daemon
	3248	3209	1000	ubuntu	ubuntu	?	0:00.08	0.0	0.0	S	pipewire
	3252	3209	1000	ubuntu	ubuntu	?	0:00.06	0.0	0.0	S	gnome-keyring-d
	3261	3209	1000	ubuntu	ubuntu	?	0:00.02	0.0	0.0	S	mpiris-proxy
	3265	3209	1000	ubuntu	ubuntu	?	0:00.34	0.0	0.1	S	wireplumber
	3266	3209	1000	ubuntu	ubuntu	?	0:00.00	0.0	0.0	S	pipewire

- htop (si está instalado)

```
ubuntu@ubuntu: ~ -- htop
~
0% 0.0% 0.6%
1% 0.0% 1.3%
2% 0.0% 0.0%
Mem 1.89G/29.4G Tasks: 122, 444 thr, 159 kthr; 2 running
Swp 0K/0K Load average: 0.32 0.19 0.23
Uptime: 00:15:37
```

Main	I/O	PID	USER	PRI	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	Command
		3802	ubuntu	20	0	6122M	393M	170M	S	0.6	1.3	0:20.10	/usr/bin/gnome-shell --mode=ubuntu
		4014	ubuntu	20	0	6122M	393M	0	S	0.6	1.3	0:02.81	/usr/bin/gnome-shell --mode=ubuntu
		7254	ubuntu	20	0	3750M	388M	0	S	0.6	1.3	0:01.22	/usr/bin/ptxixis --gapapplication-service
		7258	ubuntu	20	0	3750M	388M	0	S	0.6	1.3	0:00.49	/usr/bin/ptxixis --gapapplication-service
		7694	ubuntu	20	0	20764	5632	3804	R	0.6	0.0	0:00.08	htop
		1	root	20	0	26364	17592	11896	S	0.0	0.1	0:02.66	/sbin/init autoinstall --- splash noprompt noshell automatic-ubiquity
		996	root	19	-1	58028	19988	10200	S	0.0	0.1	0:00.61	/usr/lib/systemd/systemd-journald
		1046	systemd-oo	20	0	16804	7832	6780	S	0.0	0.0	0:00.26	/usr/lib/systemd/systemd-oomd
		1053	root	20	0	41632	15384	9364	S	0.0	0.0	0:00.40	/usr/lib/systemd/systemd-udev
		1057	systemd-re	20	0	22680	15312	12560	S	0.0	0.0	0:00.12	/usr/lib/systemd/systemd-resolved
		2044	avahi	20	0	6748	4936	4556	S	0.0	0.0	0:00.13	avahi-daemon: running [ubuntu.local]
		2045	root	20	0	2888	1932	1808	S	0.0	0.0	0:00.03	/bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
		2046	messagebus	20	0	11724	8356	4616	S	0.0	0.0	0:01.68	@dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation
		2050	root	20	0	58108	30052	14780	S	0.0	0.1	0:01.04	/usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
		2051	polkitd	20	0	302M	11272	0052	S	0.0	0.0	0:00.15	/usr/lib/polkit-1/polkitd --no-debug --log-level=notice
		2054	root	20	0	2096M	44452	26744	S	0.0	0.1	0:01.59	/snap/snapd/current/usr/lib/snapd/snapd
		2055	root	20	0	312M	8064	7280	S	0.0	0.0	0:00.12	/usr/libexec/accounts-daemon
		2057	root	20	0	10344	3112	2876	S	0.0	0.0	0:00.02	/usr/sbin/cron -f -P
		2058	root	20	0	311M	7736	6952	S	0.0	0.0	0:00.02	/usr/libexec/switcheroo-control
		2063	root	20	0	10600	9796	8372	S	0.0	0.0	0:00.17	/usr/lib/systemd/systemd-logind
		2066	root	20	0	165M	7116	5676	S	0.0	0.0	0:00.01	/usr/sbin/zed -F
		2076	avahi	20	0	6560	1740	1436	S	0.0	0.0	0:00.00	avahi-daemon: chroot helper
		2089	root	20	0	165M	7116	0	S	0.0	0.0	0:00.00	/usr/sbin/zed -f
		2090	root	20	0	165M	7116	0	S	0.0	0.0	0:00.00	/usr/sbin/zed -f
		808 MB Volume	20	0	311M	7736	0	0	S	0.0	0.0	0:00.00	/usr/libexec/switcheroo-control
		2099	root	20	0	311M	7736	0	S	0.0	0.0	0:00.00	/usr/libexec/switcheroo-control
		2111	root	20	0	530M	15628	12792	S	0.0	0.1	0:00.09	/usr/libexec/udisks2/udisksd
		2133	polkitd	20	0	302M	11272	0	S	0.0	0.0	0:00.03	/usr/lib/polkit-1/polkitd --no-debug --log-level=notice

```
F1:Main F2:Setup F3:Search F4:Filter F5:Tree F6:SortBy F7:Filter F8:Filter F9:Kill F10:Quit
```

- ps -aux

```
ubuntu@ubuntu:~$ ps -aux
USER        PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.2  0.0 26364 17592 ?        Ss   18:19   0:02 /sbin/init autoinstall --- splash noprompt noshell automatic-ubiquity
root         2  0.0  0.0      0     0 ?        S    18:19   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        S    18:19   0:00 [pool_workqueue_release]
root         4  0.0  0.0      0     0 ?        I<   18:19   0:00 [kworker/R-rcu_gp]
root         5  0.0  0.0      0     0 ?        I<   18:19   0:00 [kworker/R-sync_wq]
root         6  0.0  0.0      0     0 ?        I<   18:19   0:00 [kworker/R-kvfree_rcu_reclaim]
root         7  0.0  0.0      0     0 ?        I<   18:19   0:00 [kworker/R-slub_flushwq]
root         8  0.0  0.0      0     0 ?        I<   18:19   0:00 [kworker/R-netns]
root        10  0.0  0.0      0     0 ?        I<   18:19   0:00 [kworker/0:0H]
root        11  0.0  0.0      0     0 ?        I    18:19   0:00 [kworker/0:1-events]
root        12  0.0  0.0      0     0 ?        I    18:19   0:00 [kworker/u24:0-ipv6_addrconf]
root        13  0.0  0.0      0     0 ?        I<   18:19   0:00 [kworker/R-mm_percpu_wq]
root        14  0.0  0.0      0     0 ?        S    18:19   0:00 [ksoftirqd/0]
root        15  0.0  0.0      0     0 ?        I    18:19   0:00 [rcu_preempt]
root        16  0.0  0.0      0     0 ?        S    18:19   0:00 [rcu_exp_par_gp_kthread_worker/0]
root        17  0.0  0.0      0     0 ?        S    18:19   0:00 [rcu_exp_gp_kthread_worker]
root        18  0.0  0.0      0     0 ?        S    18:19   0:00 [migration/0]
root        19  0.0  0.0      0     0 ?        S    18:19   0:00 [kprobe-optimizer]
root        20  0.0  0.0      0     0 ?        S    18:19   0:00 [idle_inject/0]
root        21  0.0  0.0      0     0 ?        S    18:19   0:00 [cpuhp/0]
root        22  0.0  0.0      0     0 ?        S    18:19   0:00 [cpuhp/1]
root        23  0.0  0.0      0     0 ?        S    18:19   0:00 [idle_inject/1]
root        24  0.0  0.0      0     0 ?        S    18:19   0:00 [migration/1]
root        25  0.0  0.0      0     0 ?        S    18:19   0:00 [ksoftirqd/1]
root        27  0.2  0.0      0     0 ?        I<   18:19   0:01 [kworker/1:0H-kblockd]
root        28  0.0  0.0      0     0 ?        S    18:19   0:00 [cpuhp/2]
root        29  0.0  0.0      0     0 ?        S    18:19   0:00 [idle_inject/2]
root        30  0.0  0.0      0     0 ?        S    18:19   0:00 [migration/2]
root        31  0.0  0.0      0     0 ?        S    18:19   0:00 [ksoftirqd/2]
root        33  0.0  0.0      0     0 ?        I<   18:19   0:00 [kworker/2:0H-kblockd]
root        34  0.0  0.0      0     0 ?        S    18:19   0:00 [cpuhp/3]
root        35  0.0  0.0      0     0 ?        S    18:19   0:00 [idle_inject/3]
root        36  0.0  0.0      0     0 ?        S    18:19   0:00 [migration/3]
root        37  0.0  0.0      0     0 ?        S    18:19   0:00 [ksoftirqd/3]
root        38  0.0  0.0      0     0 ?        I    18:19   0:00 [kworker/3:0-events]
root        39  0.0  0.0      0     0 ?        I<   18:19   0:00 [kworker/3:0H-kblockd]
```

ACTIVIDAD 2: Simulando prioridades (nice)

Objetivo: Entender planificación por prioridades.

Ejecución: Abrir terminal y correr

- nice -n 10 yes > /dev/null & nice -n -5 yes > /dev/null &

```
ubuntu@ubuntu:~$ nice -n 10 yes > /dev/null & sudo nice -n -5 yes > /dev/null &
[1] 7895
[2] 7896
```

ACTIVIDAD 3: Round Robin en Linux (conceptual + práctica)

Objetivo: Relacionar Linux con Round Robin.

Ejercicio:

1. Ejecutar varios procesos:

- yes > /dev/null & yes > /dev/null & yes > /dev/null &

```
ubuntu@ubuntu:~$ yes > /dev/null &
[1] 6662
ubuntu@ubuntu:~$ yes > /dev/null &
[2] 6664
ubuntu@ubuntu:~$ yes > /dev/null &
[3] 6666
```

2. Luego observar con:

- top

```
ubuntu@ubuntu:~$ top
top - 13:55:30 up 26 min, 1 user, load average: 2.31, 0.88, 0.56
Tasks: 255 total, 4 running, 251 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.3 us, 0.7 sy, 0.0 ni, 99.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7746.1 total, 267.8 free, 2401.9 used, 5666.2 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 5344.2 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
6664	ubuntu	20	0	16112	7104	5868	R	100.0	0.1	1:34.93	yes
6666	ubuntu	20	0	16112	7376	6140	R	100.0	0.1	1:21.52	yes
6662	ubuntu	20	0	16112	7288	6052	R	100.0	0.1	1:48.71	yes
6508	ubuntu	20	0	2927428	400988	136108	S	2.0	5.1	0:05.04	ptxix
3653	ubuntu	20	0	5137416	372176	163284	S	1.3	4.7	0:52.87	gnome-s+
1726	message+	20	0	11732	8476	4752	S	0.3	0.1	0:02.31	dbus-da+
1	root	20	0	26360	17548	11752	S	0.0	0.2	0:05.03	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.01	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_wor
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
9	root	20	0	0	0	0	T	0.0	0.0	0:00.00	kworker+

ACTIVIDAD 4: Cambiando prioridades en vivo

Objetivo: Ver cómo afecta la planificación.

Ejecución

Ejecutar proceso:

- `yes > /dev/null &`
- `ps -aux | grep yes` (Obtener PID)
- `renice -n 15 -p PID` (Cambiar prioridad)

```
ubuntu@ubuntu:~$ yes > /dev/null &
[1] 6612
ubuntu@ubuntu:~$ ps -aux | grep yes
ubuntu      6612  99.9  0.0 16112  7092 pts/0    R   13:34   1:51 yes
ubuntu      6617   0.0  0.0 18408  2388 pts/0    S+  13:36   0:00 grep --color=
auto yes
ubuntu@ubuntu:~$ renice -n 15 -p 6612
6612 (process ID) old priority 0, new priority 15
ubuntu@ubuntu:~$ kill 6612
[1]+  Terminated                  yes > /dev/null
ubuntu@ubuntu:~$
```

ACTIVIDAD 5: Identificando tipos de procesos

Objetivo: Diferenciar CPU-bound vs I/O-bound

Ejercicio

Ejecutar:

- `yes > /dev/null & sleep 100 &`

```
ubuntu@ubuntu: ~  
ubuntu@ubuntu:~$ yes>/dev/null &  
[1] 6462  
ubuntu@ubuntu:~$ sleep 100 &  
[2] 6469  
ubuntu@ubuntu:~$ ps -aux | grep yes  
ubuntu      6462  99.9   0.0   16112   7152 pts/0    R    13:34    2:26  yes  
ubuntu      6523   0.0   0.0   18408   2436 pts/0    S+   13:36    0:00  grep --color=  
auto  yes  
[2]+ Done  
ubuntu@ubuntu:~$ ps -aux |grep sleep  
ubuntu      6526   0.0   0.0   18408   2472 pts/0    S+   13:36    0:00  grep --color=  
auto  sleep  
ubuntu@ubuntu:~$
```

ACTIVIDAD 6: Ejercicio

Ejercicio:

Ejecuta:

- `yes > /dev/null & yes > /dev/null & nice -n 10 yes > /dev/null &`

```
daniel-oliva@daniel-oliva-VirtualBox:~$ yes > /dev/null &  
[1] 6248  
daniel-oliva@daniel-oliva-VirtualBox:~$ yes > /dev/null &  
[2] 6254  
daniel-oliva@daniel-oliva-VirtualBox:~$ nice -n 10 yes > /dev/null &  
[3] 6256
```